

MODEL QUESTION PAPER

DESIGN OF AUTOMOTIVE COMPONENTS

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

Module Outcome Cognitive level

1	Define factor of safety.	M 1.01	U
2	Name any two inversions of double slider crank chain.	M 1.06	U
3	What type of stresses are induced in shafts?	M 2.02	A
4	What is the primary function of bearings in a machine?	M 2.12	A
5	State two functions of couplings.	M 2.10	U
6	The effect of is to decrease the speed of belt on the driving shaft and to decrease the speed on the driven shaft.	M 3.02	U
7	What is meant by velocity ratio of gear drives?	M 3.06	U
8	List the two main consideration for the designing of an IC engine piston head.	M 4.06	U
9	What is the function of connecting rod of an internal combustion engine?	M4.08	U

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

Module Outcome Cognitive level

1	Differentiate between lower pair and higher pair.	M 1.03	U
2	Differentiate between machine mechanism and structure.	M 1.03	U
3	What are the factors to be considered while designing a shaft?	M 2.02	A
4	What are the requirements of a good coupling?	M 2.10	U
5	List the major classification of bearings.	M 2.12	A

6	How do you differentiate between a coupling and a clutch?	M 3.01	A
7	A shaft running at 120 rpm is driving another parallel shaft having 600mm diameter pulley. The pulley on the driving shaft has its outer diameter 800mm. Find the speed of the driven shaft.	M 3.09	A
8	Differentiate between radial follower and offset follower.	M 4.01	U
9	State the function of the following a) piston skirt b) piston pin	M 4.06	U
10	What is the role of piston in an internal combustion engine?	M 4.06	U

PART C

III. Answer all questions from the following

(6 x 7 = 42 Marks)

Module Outcome Cognitive level

III	Describe the general design procedures for machine elements.	M 1.01	U
	OR		
IV	Explain crank and slotted lever quick return mechanism with the aid of a neat sketch.	M 1.05	U
V	A hollow shaft having an inside diameter 60% of its outside diameter and has to transmit 180kW at 90rpm. If the shear stress is not to exceed 60MPa, find diameters of the shaft.	M 2.03	A
	OR		
VI	Design a muff coupling to connect two shafts transmitting 120kW at 210rpm. The permissible shearing and crushing stresses for shaft and key materials are 60MPa and 120MPa respectively. The material of the muff is cast iron with permissible shear stress of 25MPa.	M 2.11	A

VII	A line shaft is to transmit 50kW at 150rpm. The shaft pieces are connected by muff coupling. Design the coupling assuming an overload of 20 percent. The following permissible stresses may be used. Shear stress for shaft and key material = 40MPa Crushing stress for key material = 80MPa Shear stress for muff material = 15MPa OR	M 2.11	A
VIII	The standard cross section for a flat key which is fitted on a 60mm diameter shaft is 16mm x 10mm. The key is transmitting 540Nm torque from the shaft to the hub. The shearing and crushing stresses for the key material are 40MPa and 76MPa. Determine the length of the key	M 2.09	A
IX	The outer and inner diameters of a single plate clutch having friction linings on both sides are 300mm and 150mm respectively. The coefficient of friction between the contacting surfaces is 0.35. Assuming uniform rate of wear, determine the maximum power that can be transmitted by the clutch at 2000rpm when the maximum pressure is not to exceed 150kPa OR	M 3.03	A
X	A multiplate disc clutch transmits 23kW of power at 1575 rpm. Coefficient of friction for the friction surface is 0.3. Axial intensity of pressure is not to exceed 100kPa. The internal radius of the contact surface is 60mm and is 0.6 times the external radius. Find the number of plates needed to transmit the required torque. Assume uniform wear.	M 3.05	A
XI	A 20 teeth spur gear A is rotating at 40rpm in clockwise direction. Find out the speed and direction of rotation of another 45 teeth external spur gear B rotating with gear A. OR	M 3.07	A
XII	A shaft runs at 450 rpm and drives another shaft at 650rpm through belt drive. The diameter of the driving pulley is 500mm. Determine the diameter of the driven pulley in the following cases. (i) Neglecting belt thickness (ii) Taking belt thickness as 5mm (iii) Assuming a total slip of 5 percentage	M 3.10	A

XIII	Draw the profile of a cam to give the following motion to a knife edged follower. 1)outstroke during 60 degree of cam rotation 2)dwel for the next 30 degree of cam rotation 3)return stroke during next 60 degree of cam rotation 4)dwel for the remaining rotation of the cam The stroke of the follower is 20mm and the minimum radius of cam is 50mm.The axis of the follower is offset by 20mm from the axis of the cam. The follower moves with uniform velocity during both the outstroke and return stroke OR	M 4.03	A
XIV	List the seven important design procedures for an IC engine connecting rod.	M 4.08	A